Discrete Morse Theory

Anamitro Biswas
Webpage: https://anamitro.github.io

September 27, 2025

1 Simplicial complexes

1.1 Affine independence

1.2 Geometric simplicial complex

1.3 Abstract simplicial complex

1.4 Equivalence

1.5 Simplicial maps

2 Collapsing

3 Discrete Morse functions

For K an abstract simplicial complex, a map $f : K \to \mathbb{R}$ is called a *discrete Morse function* if $\forall\, \sigma^k \in K$

(i) $e_1 = \left|\left\{\tau^{k-1} \in K \mid f(\tau) \geq f(\sigma)\right\}\right| \leq 1$;

(ii) $e_2 = \left|\left\{\tau^{k+1} \in K \mid f(\tau) \leq f(\sigma)\right\}\right| \leq 1$.

Every function g that *respects dimension* in the sense that for each $\sigma^{k-1} \subset \tau^k \in K$, $g(\sigma) < g(\tau)$ is a DMF. On the other hand, a DMF *almost* respects dimension by ensuring that each simplex can have one "exceptional" facet and one "exceptional" co-face; and in fact, not both. To arrive at a contradiction by assuming that for a face $\sigma \cup \left\{v_1\right\} \in K$, where $\sigma \in K$ and $v_1, v_2 \in K_0$, $f(\sigma) \geq f\left(\sigma \cup \left\{v_1\right\}\right) \geq f\left(\sigma \cup \left\{v_1, v_2\right\}\right)$, we have for $\sigma \cup \left\{v_2\right\} \in K$, $f\left(\sigma \cup \left\{v_1, v_2\right\}\right) > f\left(\sigma \cup \left\{v_2\right\}\right) > f(\sigma)$, the first strict inequality since the exceptional coface of σ is $\sigma \cup \left\{v_1\right\}$ and the second one since the exceptional face of $\sigma \cup \left\{v_1, v_2\right\}$ is $\sigma \cup \left\{v_1\right\}$. Simplices with exceptions form disjoint pairs of a simplex τ and its facet σ, termed *regular pairs*. It encodes an arrow $\sigma \to \tau$ in the sense that σ can be collapsed onto τ. Consecutive collapsings end up in a simplex which can't be collapsed any more, or a simplex without any exception, called a *critical simplex*.

If n_i is the number of critical simplices of dimension i in K, $\chi = \sum_{i=0}^{\infty} (-1)^i\, n_i$.

A *discrete vector field* is a disjoint collection of such pairs $\left(\sigma_i, \tau_i\right)$ where for each i, σ_i is a facet of τ_i. Each pair represents as an arrow $\to_i$. The disjointness means that each simplex can be a member of at most one pair in a vector field.

If a discrete vector field $\left\{\left(\sigma_i, \tau_i\right)\right\}_{i \in I}$ is given on a simplicial complex K, and $p \in \mathbb{N}$, a *p-path* is a sequence

$$\sigma_{i_1}^{p-1} \to \tau_{i_1}^{p} \geq \sigma_{i_2}^{p-1} \to \tau_{i_2}^{p} \geq \sigma_{i_3}^{p-1} \ldots \to \tau_{i_k}^{p} \geq \sigma_{i_{k+1}}^{p-1},$$

each arrow in the discrete vector field, and σ_{i_j} is a facet of $\tau_{i_{j-1}}$. It's a *cycle* is $\sigma_1 = \sigma_{k+1},\ k \geq 1$, and *acyclic* if it admits no cycle. A critical simplex can only appear as the last simplex in any p-path in a discrete vector field.

Regular pairs of a DMF form a discrete vector field. A discrete vector field formed by the regular pairs of a DMF is called a *gradient vector field*.

Theorem 1. *Each gradient vector field is acyclic. On the other hand, each acyclic discrete vector field on a simplicial complex K is induced by some DMF, i.e., is a gradient vector field.*

Proof. Given a DMF, in the induced discrete vector field, function values decrease along any p-path, since $f\left(\sigma_{i_j}\right) \geq f\left(\tau_{i_j}\right)$ being the only exception for $f\left(\tau_{i_j}\right)$, $f\left(\tau_{i_j}\right) > f\left(\sigma_{i_{j+1}}\right)$ for all j. Thus, *each gradient vector field is acyclic*. COMPLETE □

However, different DMF's on a simplicial complex might induce the same DVF, e.g., if f is a DMF, then so are e^f and $3f$, and all three induce the same DVF. Our primary interest in DVF's lies in their encodings of collapses and deformations-simplifications of a simplicial complex. A DMF represents a convenient but not unique way of encoding a DVF.

Proposition 2. *Suppose the critical simplices of an acyclic DVF on K form a subcomplex $L \leq K$. Then $\exists$ a collapse $K \to L$, and so, $K \simeq L$.*

For $\sigma \in K \setminus L$, we claim the existence of a regular pair (σ, τ) where σ is a *free face* of τ, whence we can perform an elementary collapse, and proceed using the same claim on the resulting complex. To prove the claim, let n denote the maximal dimension of a simplex in $K \setminus L$. There exist n-path(s) in the DVF (of which we take the maximal), and $\sigma \to \tau$ be its first regular pair. Now, $\sigma \leq \tau$, and if σ were a facet of some other simplex τ' in $K \setminus L$, then we can extend the n-path as $\tau' \geq \sigma \to \tau \geq \dots$, contradicting its maximality. On the other hand, if σ were a facet of some simplex τ' in L, then $\sigma \in L$ as L is a subcomplex, a contradiction. □

4 Morse homology

An *oriented p-path* from σ_1^{p-1} to σ_{k+1}^{p-1} is a p-path consisting of oriented simplices, such that for each j the orientation induced by τ_j on its facets (i) does not match σ_j and (ii) matches σ_{j+1}. One could say that the simplices τ_{i_j} in an oriented p-path are oriented consistently along the path.

Let us start with a simplicial complex K, an gradient vector field on it, and $\mathbb{F}$ an (algebraic) field to provide efficient algebraic constructions. For each i, let n_i denote # critical i-simplices. For $p \in \mathbb{N} \cup \{0\}$, a *Morse p-chain* is a formal sum $\sum_{i=1}^{n_p} \lambda_i \sigma_i^p$; $\lambda_i \in \mathbb{F}$, σ_i^p an oriented critical p-simplex in K for each i. The module of all such Morse p-chains is the *p-dimensional Morse chain group C_p* $\cong \mathbb{F}^{n_p}$. Now, we have to define the boundary map. Given an oriented critical p-simplex τ, let $\delta(\tau)$ be its collection of facets with induced orientation. For each $\sigma \in C_{p-1}$, let $\alpha_{\tau,\sigma} := \sum_{\sigma' \in \delta(\tau)} \#\left\{\text{oriented paths from } \sigma' \text{ to } \sigma\right\}$. Since we are considering oriented simplices, $\alpha_{\tau,\sigma}$ counts paths different from those $\alpha_{\tau,-\sigma}$ counts. For $\tau \in C_p$, we define $\partial_p \tau := \sum_{i=1}^{n_{p-1}} \left(\alpha_{\tau,\sigma_i} - \alpha_{\tau,-\sigma_i}\right) \sigma_i$. The oriented paths constituting the boundary map model how arrows stretch the boundary of a critical p-simplex towards critical $(p-1)$-simplices. As we have got a Morse chain complex $\dots \xrightarrow{\partial} C_n \xrightarrow{\partial} C_{n-1} \xrightarrow{\partial} \dots \xrightarrow{\partial} C_1 \xrightarrow{\partial} C_0 \xrightarrow{\partial} 0$ with $\partial^2 = 0$, we can define *Morse homology* of a gradient vector field on K as $\mathcal{H}(K;\mathbb{F}) = \ker \partial_p / \operatorname{im} \partial_{p+1}$. This is isomorphic to standard simplical and singular homologies. One way to see this is that nowhere do we change the homotopy type of the complex.

Weak Morse inequalities. For each p, # critical p-simplices is greater than or equal to the corresponding Betti number: $n_p \geq \mathfrak{b}_p$.

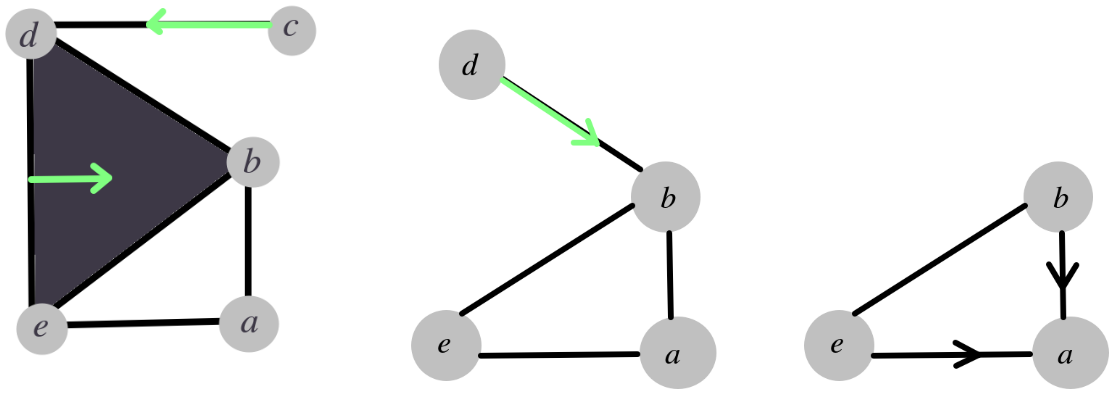
[Proof]

5 Computations

5.1 An easy complex

I have used the same complex mentioned as Example 11.3.8 in [V24].

5.1.1 Collapsings up to critical simplex



5.1.2 Morse chains

There is a critical vertex $\langle a \rangle$ and a critical edge $\langle b, e \rangle$. Thus $C_0 \cong C_1 \cong \mathbb{F}$, and other Morse chains are trivial.

5.1.3 Oriented paths

Now to calculate the boundary maps, we have $\delta(\langle b, e \rangle) = \{\langle e \rangle, -\langle b \rangle\}$, as per induced orientation, and one oriented 1-path from $\langle e \rangle$ to $\langle a \rangle$, giving the collapse $\langle e \rangle \to \langle e, a \rangle \geq \langle a \rangle$. Similarly, there is one oriented 1-path from $-\langle b \rangle$ to $-\langle a \rangle$.

5.1.4 Boundary maps

This gives us $\alpha_{\langle b,e \rangle, \langle a \rangle} = \alpha_{\langle b,e \rangle, -\langle a \rangle} = 1$, and hence $\partial \langle b, e \rangle = \left(\alpha_{\langle b,e \rangle, \langle a \rangle} - \alpha_{\langle b,e \rangle, -\langle a \rangle}\right) . \langle a \rangle = 0$.

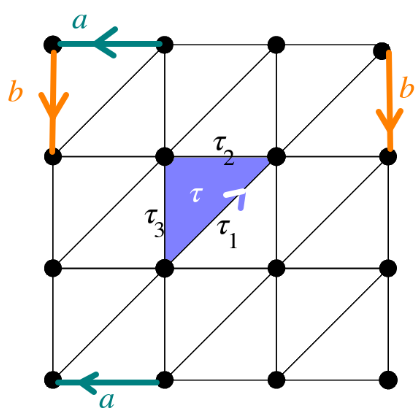
5.1.5 Chain complex and homology

$$\ldots \to 0 \xrightarrow{\partial_2 = \mathrm{id}_0} \mathbb{F} \xrightarrow{\partial_1 = 0} \mathbb{F} \xrightarrow{\partial_0 = 0} \mathbb{F}$$

So, $\mathcal{H}_0(K) \cong \mathcal{H}_1(K) \cong \mathbb{F}$, and higher homology groups are trivial.

5.2 Torus

We have that a torus can be formed from a rectangle by identifying opposite edges in the same orientation. However, this is the more formal way of triangulating a torus (as a simplicial complex, not as a CW complex), because of the following reasons:



(i) In a simplicial complex, all the faces of a simplex are required to be different, because if a simplex is embedded in a real vector space, all its faces are different; but there a, b and the diagonal would not be embedded (mapped injectively) since each has two endpoints attached to the same vertex.

(ii) Intersection of any pair of simplices is required to be a simplex, but in a rectangle, two triangular simplices intersect at the union of the sides: not a simplex.

5.2.1 Triangulation of a torus

5.2.2 Collapsing and Morse chain groups

It is easy to check that the torus can be serially collapsed on to the simplex colored green. Let's call it τ, which is homotopic to the entire torus. We see that x is a critical vertex, a and b are critical edges, and τ is the critical triangle. The Morse chain complex is of the form $\ldots \to 0 \to \mathbb{F} \to \mathbb{F}^2 \to \mathbb{F} \to 0$. We need now to compute the boundary function.

5.2.3 Boundary functions and homology

We see that $\delta(\tau)$ has 3 edges, which we name as τ_1, τ_2 and τ_3 as in the figure. From τ_1, there is one oriented 2-path to critical edge $-\langle a \rangle$ (marked in blue) and another oriented 2-path to critical edge $-\langle b \rangle$ (marked in red). From τ_2, we have such two oriented 2-paths to $\langle a \rangle$ and $\langle b \rangle$ respectively marked in green and yellow. No oriented 2-path from τ_3 end up in a critical edge. Thus, $\partial(\tau) = -\langle a \rangle - \langle b \rangle + \langle a \rangle + \langle b \rangle = 0$. Similarly, $\partial a = \partial b = 0$, so $\partial_1 \equiv 0$. This gives the Morse chain complex:

$$\ldots \to 0 \to \mathbb{F} \xrightarrow{\partial_2 \equiv 0} \mathbb{F}^2 \xrightarrow{\partial_1 \equiv 0} \mathbb{F} \to 0.$$

Thus, $\mathcal{H}_0(K) \cong \mathcal{H}_2(K) \cong \mathbb{F}$ and $\mathcal{H}_1(K) \cong \mathbb{F}^2$.

6 Grid graphs

7 Computation with grid graph

References

[V24] Ž. Virk, *Introduction to Persistent Homology*, ed. 1.03, Faculty of Computer and Information Science, University of Ljubljana (2024).